SLIIT / FACULTY OF COMPUTING

CONTEXT-AWARE IoT-BASED FIELD MONITORING

Sri Lankan Paddy Cultivation Decision Support System

PROJECT ID R26-IT-066 FINAL RESEARCH PRESENTATION

From field data to cultivation decisions

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RESEARCH CONTEXT

The real-world problem

INCOME & PRODUCTIVITY RISK

Paddy-farming households face instability linked to weather, soil imbalances and fungal diseases.

INEFFICIENT DECISIONS

Traditional knowledge and fixed guidelines can delay fertilizer timing and disease response.

YIELD-LOSS RISK

Rice Blast, Brown Spot and uninformed crop selection are identified in PP2 as potential contributors to yield loss.

FRAGMENTED TOOLS

PP2 identifies gaps between IoT monitoring, disease severity analysis and market-aware crop recommendations.



Four gaps that shape the project

01

Fragmented solutions

No single platform connects IoT, climate, disease and market intelligence.

02

Real-time IoT integration

Raw sensor readings need actionable, localized recommendations.

03

Disease severity & economic impact

Severity grading and yield-loss forecasting are identified as missing capabilities.

04

Market-aware crop selection

Recommendations should consider price fluctuations and supply–demand.

Research motivation: bring the four decision dimensions into one farmer-centric system.

RESEARCH OBJECTIVES

Develop an integrated, context-aware IoT-based decision support system for data-driven paddy cultivation decisions.

IoT field monitoring

Monitor soil moisture, pH, NPK and temperature; generate alerts.

Crop recommendation

Use agricultural, climate and market information where implemented.

Disease detection & severity

Detect supported disease classes and assess severity where implemented.

Harvest yield estimation

Estimate yield under verified scenarios to inform decisions.

PROPOSED SOLUTION

A connected decision-support journey

01

FIELD MONITORING

Capture field conditions

02

INTELLIGENT ANALYSIS

Interpret agricultural data

03

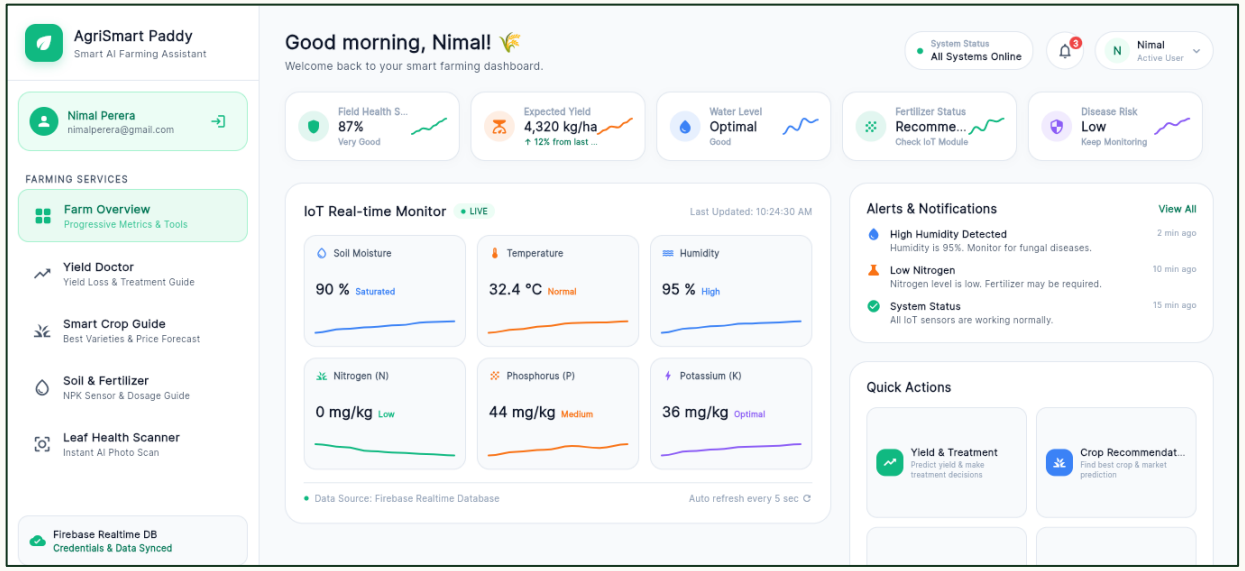
SCENARIO RECOMMENDATIONS

Support crop, disease and yield decisions

04

FARMER DASHBOARD

Present useful outputs





Four components & One cultivation workflow.

01

IoT Field Monitoring

Soil sensing and fertilization timing alerts.

Student ID: IT22178718

02

Crop Recommendation

IoT, climate and market inputs where implemented.

Student ID: IT21811104

03

Disease Detection & Severity

Paddy leaf image analysis and severity outputs where implemented.

Student ID: IT22261632

04

Harvest Yield Estimation

Field data and disease information for yield scenarios where implemented.

Student ID: IT22266446

How information moves through the system

DATA SOURCES

IoT sensors • Weather APIs • Paddy leaf images • Market / yield reference data

PROCESSING

Python / ML models listed in PP2: Random Forest, XGBoost, EfficientNet-B4 CNN

INTEGRATION

FastAPI REST endpoints listed in PP2 for cross-module communication

OUTPUT

Flutter web dashboard listed in PP2: alerts, recommendations, disease and yield outputs



STORAGE

PostgreSQL and AWS S3 are listed in PP2; verify final implementation

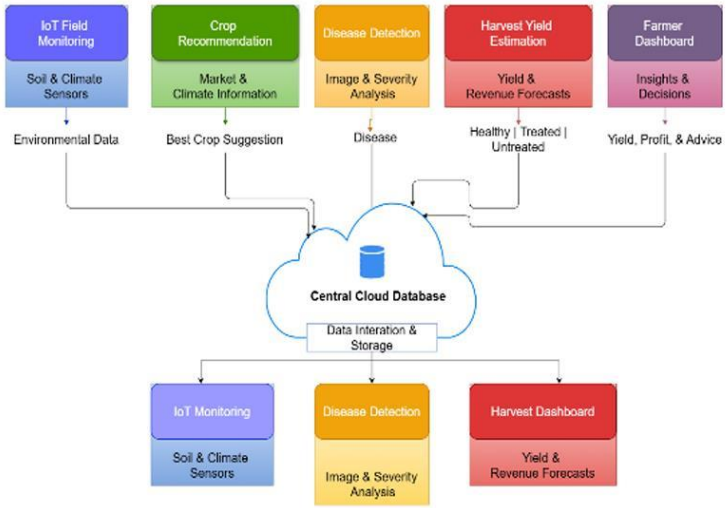
How the components connect

IoT → Crop
Soil and climate context can inform agronomic suitability.

IoT + Disease → Yield
Environmental conditions and disease severity can inform yield estimation.

Disease → Treatment
Severity and yield impact can support treatment decisions where implemented.

All components → Dashboard
Outputs are intended to appear in a unified interface.



From research gap to integrated validation

01

Problem & literature review

Review scholarly sources and Sri Lankan platforms.

02

Research gap & objectives

Define component goals and measurable outcomes.

03

System design

Design architecture and shared data flow.

04

Component development

Implement individual component workflows.

05

Integration & testing

Test API integration and cross-module data flow.

06

Evaluation & validation

Evaluate component metrics and overall functionality.

The data and technology landscape

DATA SOURCES

- IoT sensor data: soil moisture, pH, NPK, temperature and humidity
- Weather APIs and climate forecasts
- Paddy leaf images
- Market-price databases
- IRRI yield-loss references and DoA guidelines

TECHNOLOGY STACK LISTED IN PP2

Backend
Python, FastAPI, AWS IoT Core / Analytics

Machine learning
Random Forest, XGBoost, EfficientNet-B4 CNN

Frontend
Flutter web dashboard

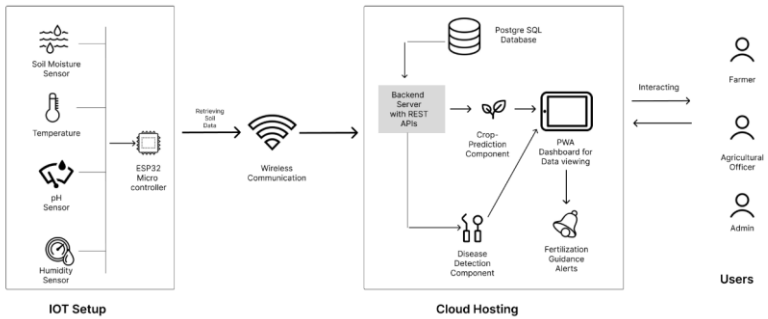
Storage / cloud
PostgreSQL, AWS S3

IoT field monitoring

Student ID: IT22178718 – Karalliyadda K.K.U.B.M.U.B

COMPONENT ROLE

Monitor field parameters and support fertilization timing alerts as described in PP2.



VERIFIED ACTION

Connectivity issue fixed; team reports the device now works well.



Crop recommendation

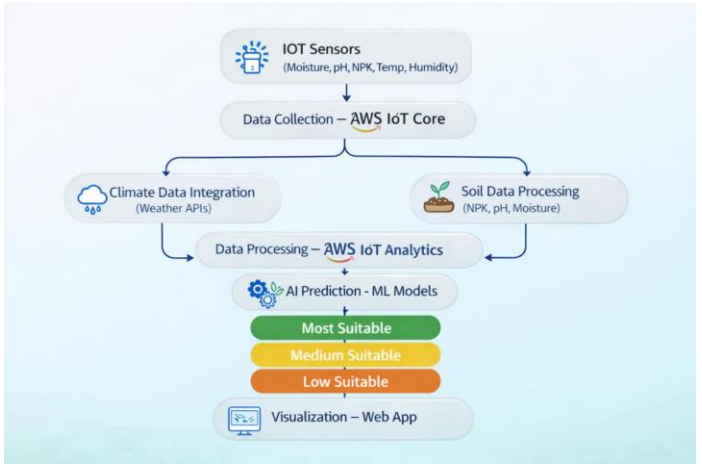
Student ID: IT21811104 – Bandara R.M.A.S.

COMPONENT ROLE

Recommend crops using agricultural, climate and market context where implemented.

WHAT TO SHOW

Actual inputs, recommendation output and test evidence.



Disease detection & severity analysis

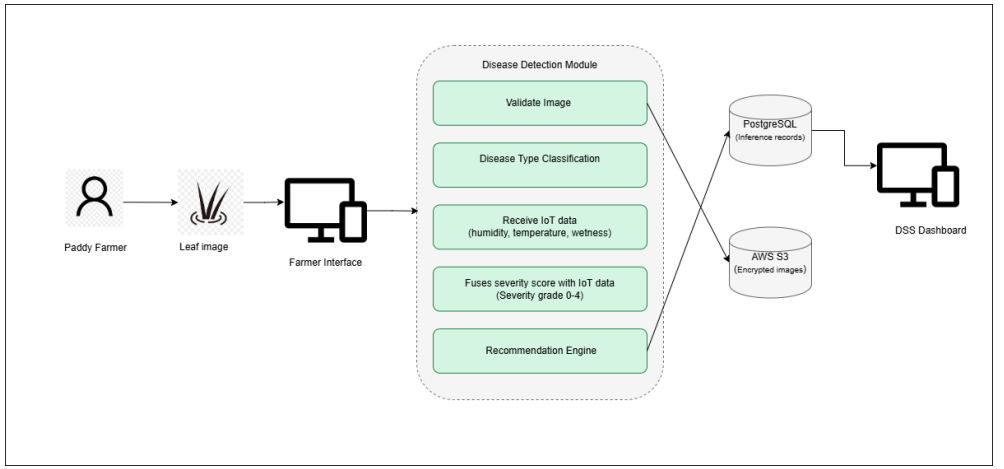
Student ID: IT22261632 – Harshika M.A.B.

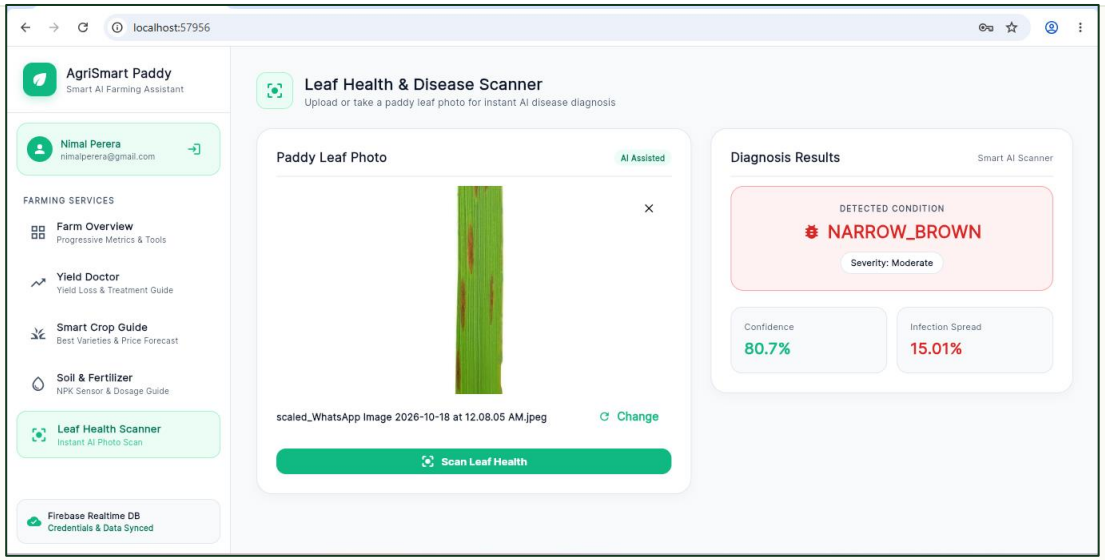
COMPONENT ROLE

Detect supported paddy disease classes and assess severity according to the implemented workflow.

WHAT TO SHOW

Actual model workflow, disease classes, predictions and measured evaluation metrics.





Harvest yield estimation & decision support

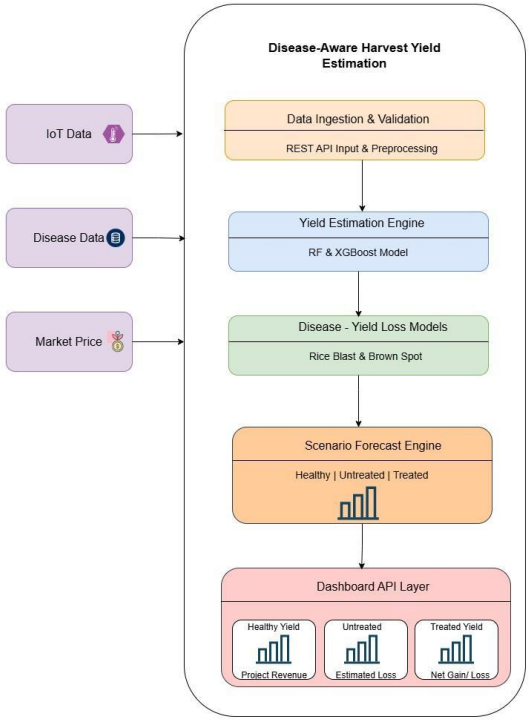
Student ID: IT22266446 - Weerasinghe W.A.H.V.

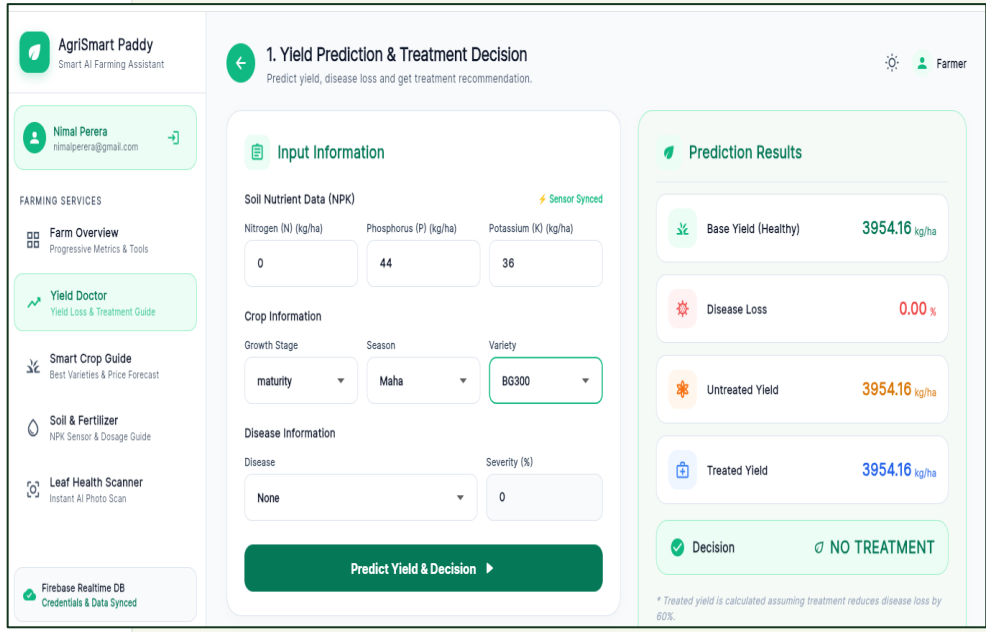
INDIVIDUAL CONTRIBUTION

Use the implemented yield-estimation workflow to support harvest and disease-aware decisions.

WHAT TO DEMONSTRATE

Actual inputs, model processing, backend/API response and final output.





PP2 FEEDBACK

Panel comments & actions taken

STUDENT / COMPONENT	PP2 PANEL COMMENT	VERIFIED ACTION / STATUS
IT22178718	IoT device had a connection problem. Moderate progress.	Connectivity issue fixed; device now works well.
IT22261632	Moderate progress for PP2 milestone.	
IT22266446	Moderate progress for PP2 milestone.	Decision logic corrected for disease status “None”.
IT21811104	Poor progress for PP2 milestone.	

CONCLUSION

What the project contributes

CONNECTED APPROACH

Four components address monitoring, crop selection, disease analysis and yield estimation.

VERIFIED FIX 1

The IoT device connection issue reported at PP2 has been fixed.

VERIFIED FIX 2

Disease status “None” now returns “No Treatment” in the yield decision logic.

LIMITATIONS

Add only limitations supported by final implementation and testing.

FUTURE WORK

Broader validation, stronger evidence and continued model refinement.



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Thank You

- Final Research Presentation

